COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 Automating Management Server	**SECTION:**	CPE31S2
DATE PERFORMED:	08-13-2026	**DATE SUBMITTED:**	08-13-2026
NAME:	Aicon H. Keliste	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. ___ PRELIM

1. PROCEDURE OUTPUT

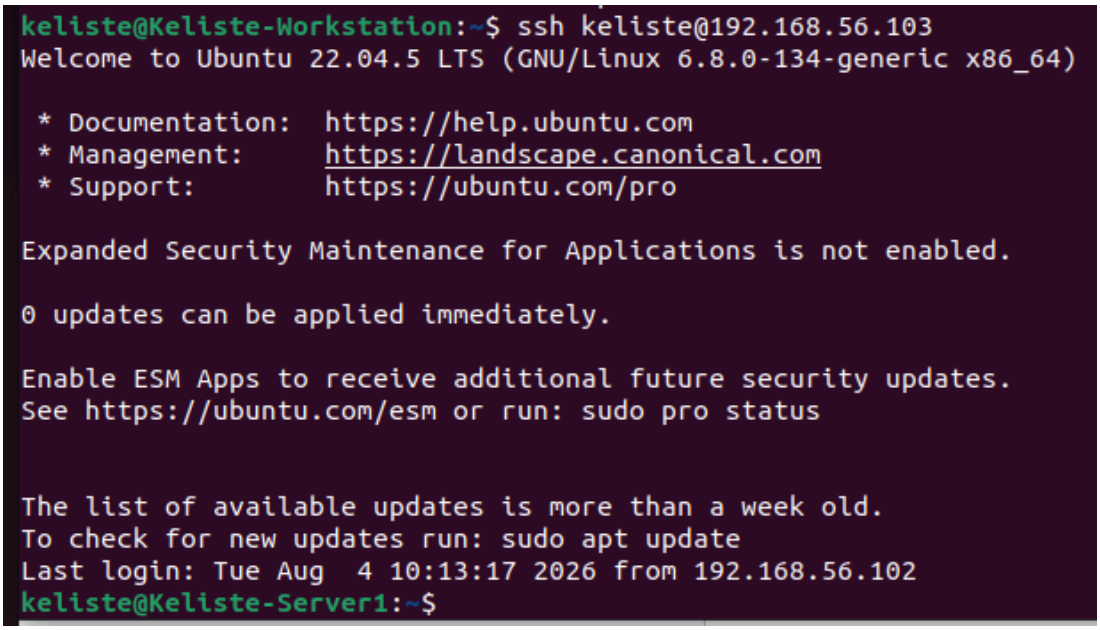
Hostname	Operating System	IP Address
keliste@Keliste-Server1	Ubuntu	192.168.56.103
keliste@Keliste-Server2	Ubuntu	192.168.56.104
keliste-centos@KelisteMain	CentOS	192.168.56.105

Part A: Installing and Verifying Ansible

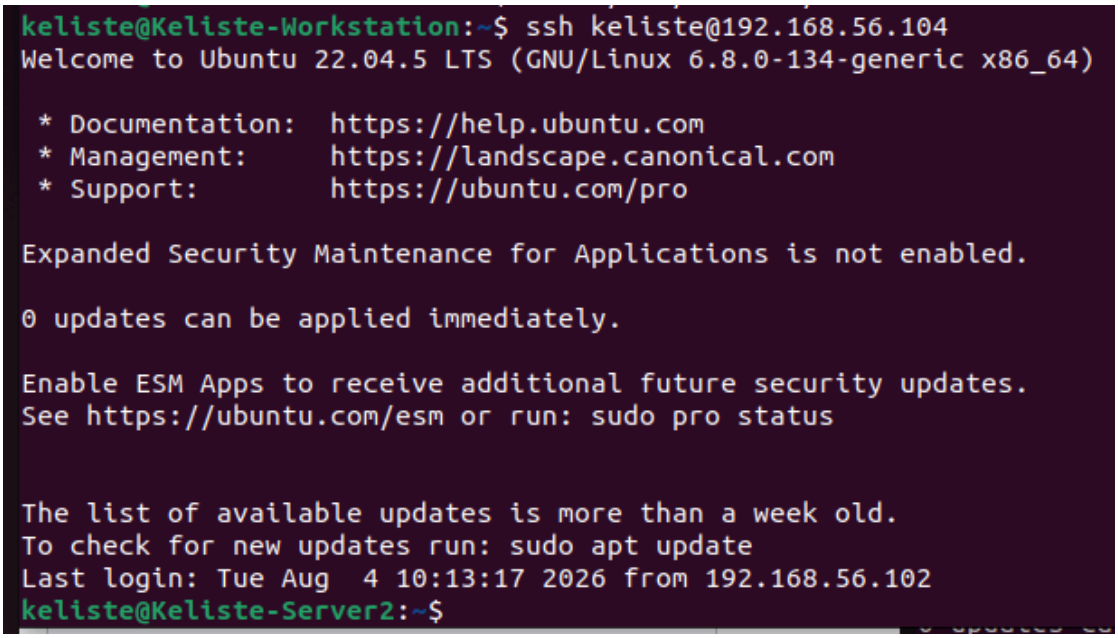
```
keliste@Keliste-Workstation:~$ sudo apt install ansible
[sudo] password for keliste:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcab-1.0-0 libsmbios-c2
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  ieee-data python-babel-localedata python3-argcomplete python3-babel
  python3-distutils python3-dnspython python3-jinja2 python3-jmespath
  python3-kerberos python3-libcloud python3-netaddr python3-ntlm-auth
  python3-packaging python3-pycryptodome python3-requests-kerberos
  python3-requests-ntlm python3-requests-toolbelt python3-selinux python3-simplejson
  python3-winrm python3-xmltodict
Suggested packages:
  cowsay sshpass python3-sniffio python3-trio python-jinja2-doc ipython3
  python-netaddr-docs
The following NEW packages will be installed:
  ansible ieee-data python-babel-localedata python3-argcomplete python3-babel
  python3-distutils python3-dnspython python3-jinja2 python3-jmespath
  python3-kerberos python3-libcloud python3-netaddr python3-ntlm-auth
  python3-packaging python3-pycryptodome python3-requests-kerberos
  python3-requests-ntlm python3-requests-toolbelt python3-selinux python3-simplejson
  python3-winrm python3-xmltodict
0 upgraded, 22 newly installed, 0 to remove and 0 not upgraded.
Need to get 28.2 MB of archives.
After this operation, 272 MB of additional disk space will be used.
Do you want to continue? [Y/n] yes
Get:1 http://ph.archive.ubuntu.com/ubuntu jammy/main amd64 python-babel-localedata all 2.8.0+dfsg.1-7 [4,982 kB]
1% [1 python-babel-localedata 206 kB/4,982 kB 4%]                    5,614 B/s 1h 23min 1s^Get:2 http://ph
.archive.ubuntu.com/ubuntu jammy/main amd64 python3-babel all 2.8.0+dfsg.1-7 [85.1 kB]
Get:3 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 python3-jinja2 all 3.0.3-1ubuntu0.4 [108 kB]
Get:4 http://ph.archive.ubuntu.com/ubuntu jammy/main amd64 python3-packaging all 21.3-1 [30.7 kB]
Get:5 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 python3-pycryptodome amd64 3.11.0+dfsg1-3ubun
tu0.1 [1,029 kB]
Get:6 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 python3-distutils all 3.10.8-1~22.04 [139 kB]
```

```
keliste@Keliste-Workstation:~$ ansible --version
ansible 2.10.8
  config file = None
  configured module search path = ['/home/keliste/.ansible/plugins/modules', '/u
sr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  executable location = /usr/bin/ansible
  python version = 3.10.12 (main, Jun 22 2026, 18:55:27) [GCC 11.4.0]
keliste@Keliste-Workstation:~$
```

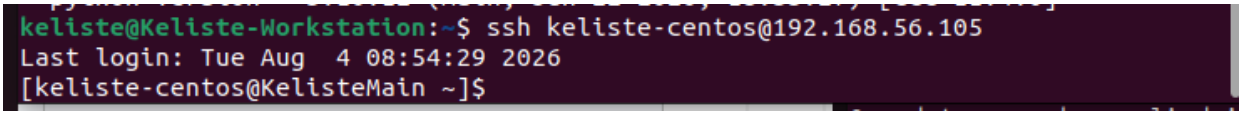
Server 1



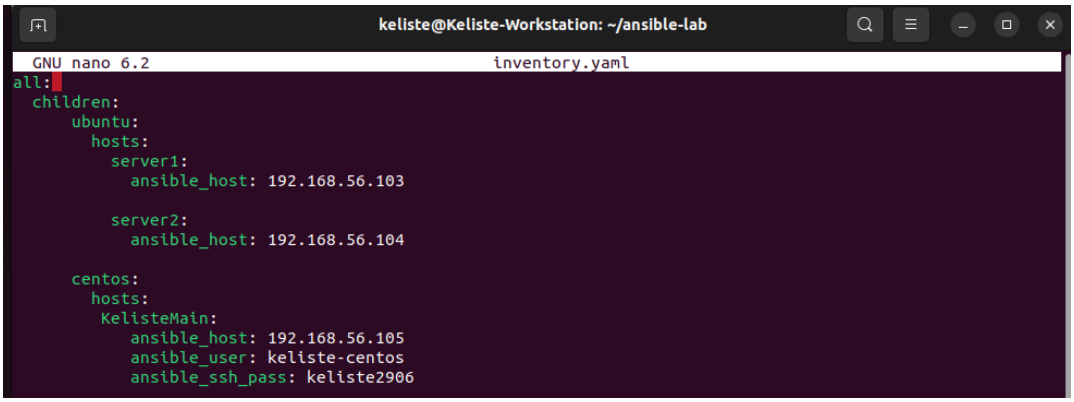
Server 2

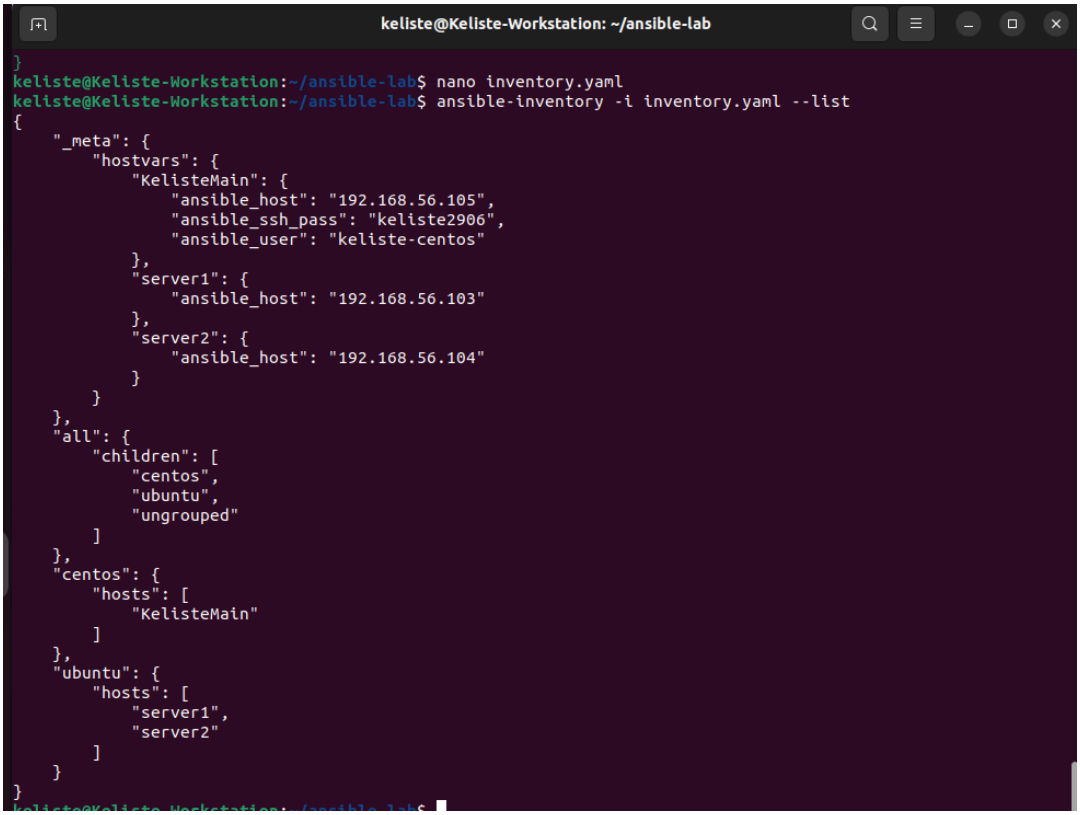


CentOS



Part B: Building an Inventory File





```
keliste@Keliste-Workstation:~/ansible-lab$ ansible all -m ping -i inventory.yaml
server2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
server1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
[DEPRECATION WARNING]: Distribution centos 9 on host KelisteMain should use /usr/libexec/platform-python,
but is using /usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible
release will default to using the discovered platform python for this host. See
https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for more
information. This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting
 deprecation_warnings=False in ansible.cfg.
KelisteMain | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python"
    },
    "changed": false,
    "ping": "pong"
}
keliste@Keliste-Workstation:~/ansible-lab$
```

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

```
keliste@Keliste-Workstation:~/ansible-lab$ ansible all --list-hosts
  hosts (3):
    192.168.56.103
    192.168.56.104
    192.168.56.105
keliste@Keliste-Workstation:~/ansible-lab$
```

Part C: Creating a Metagroup

keliste@Keliste-Workstation: ~/ansible-lab

```
 GNU nano 6.2                                 inventory.yaml *
all:
  children:
      ubuntu:
        hosts:
          server1:
            ansible_host: 192.168.56.103

          server2:
            ansible_host: 192.168.56.104

      centos:
        hosts:
         KelisteMain:
            ansible_host: 192.168.56.105
            ansible_user: keliste-centos
            ansible_ssh_pass: keliste2906

linuxservers:
  children:
    ubuntu:
    centos:
```

```
keliste@Keliste-Workstation:~/ansible-lab$ nano inventory.yaml
keliste@Keliste-Workstation:~/ansible-lab$ ansible linuxservers -m ping -i inventory.yaml
server1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
server2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
[DEPRECATION WARNING]: Distribution centos 9 on host KelisteMain should use /usr/libexec/platform-python,
but is using /usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible
release will default to using the discovered platform python for this host. See
https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for more
information. This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting
 deprecation_warnings=False in ansible.cfg.
KelisteMain | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python"
    },
    "changed": false,
    "ping": "pong"
}
keliste@Keliste-Workstation:~/ansible-lab$
```

```
keliste@Keliste-Workstation:~/ansible-lab$ ansible linuxservers --list-hosts -i inventory.yaml
  hosts (3):
    server1
    server2
    KelisteMain
keliste@Keliste-Workstation:~/ansible-lab$
```

Part D: Creating Variables

```
  GNU nano 6.2                                  inventory.yaml *
all:
  children:
      ubuntu:
        hosts:
          server1:
            ansible_host: 192.168.56.103
            os_type: Ubuntu
          server2:
            ansible_host: 192.168.56.104
            os_type: Ubuntu
      centos:
        hosts:
         KelisteMain:
            ansible_host: 192.168.56.105
            ansible_user: keliste-centos
            ansible_ssh_pass: keliste2906
            os_type: CentOS

linuxservers:
  children:
    ubuntu:
    centos:
```

```
 GNU nano 6.2                                  inventory.yaml *
all:
  children:
      ubuntu:
        vars:
          ansible_user: keliste
        hosts:
          server1:
            ansible_host: 192.168.56.103
            os_type: Ubuntu
          server2:
            ansible_host: 192.168.56.104
            os_type: Ubuntu
      centos:
        vars:
          ansible_user: keliste-centos
        hosts:
         KelisteMain:
            ansible_host: 192.168.56.105
            ansible_ssh_pass: keliste2906
            os_type: CentOS

linuxservers:
  children:
    ubuntu:
    centos:
```

keliste@Keliste-Workstation: ~/ansible-lab

```
 GNU nano 6.2                                  testvars.yaml
---
- name: Display Variables
  hosts: linuxservers

  tasks:

    - name: Show Operating System
      debug:
        msg: "Host {{ inventory_hostname }} is running {{ os_type }}"
```

```
keliste@Keliste-Workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml testvars.yaml

PLAY [Display Variables] ***************************************************************************

TASK [Gathering Facts] *****************************************************************************
ok: [server2]
[DEPRECATION WARNING]: Distribution centos 9 on host KelisteMain should use /usr/libexec/platform-python,
but is using /usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible
release will default to using the discovered platform python for this host. See
https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for more
information. This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting
 deprecation_warnings=False in ansible.cfg.
ok: [KelisteMain]
ok: [server1]

TASK [Show Operating System] ***********************************************************************
ok: [KelisteMain] => {
    "msg": "Host KelisteMain is running CentOS"
}
ok: [server2] => {
    "msg": "Host server2 is running Ubuntu"
}
ok: [server1] => {
    "msg": "Host server1 is running Ubuntu"
}

PLAY RECAP *****************************************************************************************
KelisteMain                : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ig
nored=0
server1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ig
nored=0
server2                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ig
nored=0

keliste@Keliste-Workstation:~/ansible-lab$
```

Part E: Creating Your First Playbook

```
  GNU nano 6.2                                                playbook.yaml
---
- name: First Ansible Playbook
  hosts: linuxservers

  tasks:

    - name: Ping all managed nodes
      ping:

    - name: Display Hello World
      debug:
        msg: " Hello World from {{ inventory_hostname }}"

    - name: Display Operating System
      debug:
        msg: " Operating System:  {{ os_type }}"
```

```
keliste@Keliste-Workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml playbook.yaml

PLAY [First Ansible Playbook] ***************************************************************************

TASK [Gathering Facts] **********************************************************************************
ok: [server1]
ok: [server2]
[DEPRECATION WARNING]: Distribution centos 9 on host KelisteMain should use /usr/libexec/platform-python,
but is using /usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible
release will default to using the discovered platform python for this host. See
https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for more
information. This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting
 deprecation_warnings=False in ansible.cfg.
ok: [KelisteMain]

TASK [Ping all managed nodes] ***************************************************************************
ok: [server2]
ok: [server1]
ok: [KelisteMain]

TASK [Display Hello World] ******************************************************************************
ok: [server1] => {
    "msg": " Hello World from server1"
}
ok: [server2] => {
    "msg": " Hello World from server2"
}
ok: [KelisteMain] => {
    "msg": " Hello World from KelisteMain"
}

TASK [Display Operating System] *************************************************************************
ok: [server1] => {
    "msg": " Operating System:  Ubuntu"
}
ok: [server2] => {
    "msg": " Operating System:  Ubuntu"
}
ok: [KelisteMain] => {
    "msg": " Operating System:  CentOS"
}

PLAY RECAP **********************************************************************************************
KelisteMain                : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server1                    : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2                    : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Servers

```
  GNU nano 6.2                                          inventory.yaml
all:
  children:
      ubuntu:
        vars:
          ansible_user: keliste
        hosts:
          server1:
            ansible_host: 192.168.56.103
            os_type: Ubuntu
          server2:
            ansible_host: 192.168.56.104
            os_type: Ubuntu
      centos:
        vars:
          ansible_user: keliste-centos
        hosts:
         KelisteMain:
            ansible_host: 192.168.56.105
            ansible_ssh_pass: keliste2906
            os_type: CentOS

linuxservers:
  children:
    ubuntu:
    centos:

servers:
  hosts:
    server1:
    KelisteMain:
```

Challenge.yaml

```
  GNU nano 6.2
---
- name: Display Variables
  hosts: servers

  tasks:

    - name: Display Message
      debug:
        msg: " Welcome to Ansible Laboratory "
```

```
keliste@Keliste-Workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml challenge.yaml

PLAY [Display Variables] ***************************************************************************************************

TASK [Gathering Facts] *****************************************************************************************************
ok: [server1]
[DEPRECATION WARNING]: Distribution centos 9 on host KelisteMain should use /usr/libexec/platform-python, but is using
/usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible release will default to using the
discovered platform python for this host. See
https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for more information. This feature will be
 removed in version 2.12. Deprecation warnings can be disabled by setting deprecation_warnings=False in ansible.cfg.
ok: [KelisteMain]

TASK [Display Message] *****************************************************************************************************
ok: [server1] => {
    "msg": " Welcome to Ansible Laboratory "
}
ok: [KelisteMain] => {
    "msg": " Welcome to Ansible Laboratory "
}

PLAY RECAP *****************************************************************************************************************
KelisteMain                : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

keliste@Keliste-Workstation:~/ansible-lab$
```

Guide Questions

1. What is the role of the Control Node in Ansible?
   - The Control Node is the machine where Ansible itself is installed and run from. It's where you write playbooks, inventory files, and execute commands. Also it pushes configuration out to managed nodes over SSH.
2. Why is an inventory file required?
   - Ansible needs to know which hosts to manage, how to reach them with their hostname or IP and how they're grouped.
3. What is the purpose of a metagroup?
   - The purpose of metagroup lets you organize existing groups into a larger logical group. This means you can run tasks against multiple groups at once, or apply shared variables to several groups without duplicating them.
4. How do variables simplify Ansible administration?
   - Variables let you write one generic playbook/task and reuse it across many hosts with different values.
5. What is the difference between an ad hoc command and a playbook?
   - An ad hoc command is a single, one-off Ansible command run directly from the CLI.

Conclusion:

With this prelim skills exam, I learned how to set up and troubleshoot Ansible in a real working environment. This exam demonstrated the practical setup and use of Ansible for managing multiple Linux hosts from a central control node, through configuring the inventory file, testing connectivity with the ping module, and writing a playbook to display variables.

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